Course Name: Discrete Mathematics

245460

Course Outcome

CO1: Understand the notion of mathematical thinking and proofs to solve the problem.
CO2: Apply the basics of discrete probability and number theory to solve the real world problem.
CO3: Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5. Analyze the validity of an argument using logical notation.
CO6. Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7. Understand the basic principles of sets, set equalities and operations in sets.
CO8. Apply counting principles to determine probabilities.

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Mid Term Examination, Even Semester 2025-26
B.Tech. (CS/AIML), Year-II, Semester-IV
BCSC 1010 Discrete Mathematics

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
Assume the variables to solve the numerical wherever is needed by specifying its utility.
Write the formula clearly wherever is applicable.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Let A = {1, 2, 3, 4}. Define a relation R on A as: $(a,b)\in R \Leftrightarrow a \equiv b \pmod 2$ Show that (R) is an equivalence relation.	3	4	A	F
2	A) How many different arrangements can be made from the letters of the word **ARRANGEMENT**? B) If a drawer contains socks of 4 different colors, prove that selecting 9 socks guarantees at least 3 socks of the same color.	3	2	A	F
3	Prove by mathematical induction that $7^n - 1$ is divisible by 6 for all $n\in N$.	3	6	U	C
4	Let A, B, and C be sets. Show that $(A\cup(B\cap C))' = (C' \cup B') \cap A'$.	3	6	U	C
5	Solve the recurrence relation with initial condition $a_r = 3a_{r-1} + 2, \quad r \geq 1, \quad a_0 = 1$	3	2	R	F

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Solve the recurrence relation using generating functions: $a_n = a_{n-1} + 2$ for $n \geq 1$, given that $a_0 = 1$.	5	3	A	C

7	Consider the set A={1,2,3,6} with the relation R defined as: a R b if and only if a divides b. 1. Prove that (A, R) is a partially ordered set (poset). 2. Draw the Hasse diagram of the poset. 3. Determine whether this poset forms a lattice. If yes, find the least upper bound (LUB) and greatest lower bound (GLB) of the elements 2 and 3.	5	4	An	P
8	Out of 250 candidates who failed in an examination, it was found that 128 candidates failed in Java, 87 candidates failed in C++, and 134 candidates failed in Python.Among these candidates, 31 failed in both Java and C++, 54 failed in both Java and Python, and 30 failed in both Python and C++. Using the Principle of Inclusion and Exclusion, find the number of candidates those who failed in C++ but not in Java or Python.	5	4	An	P